Who benefits most?

Sorted partial effects · A colon cancer trial, R code and inference

Valentin Klotzbücher · Department of Clinical Research

Session overview

1. Method, software and a constructed example.
2. A public colon cancer trial.
3. Sorted effects and characteristics of the tails.
4. Inference, validation and discussion.

Method and motivation

01

The average effect is useful. What else can we learn from the same model?

Two starting points

- **Applied tutorial:** *The R Journal* (Chen et al. 2020).
- **Method:** *Econometrica* (Chernozhukov, Fernández-Val, and Luo 2018).

Average effect

How large is the effect on average?

Sorted effects

How much do the fitted effects vary?

Classification

Who is in the low and high tails?

What the R package provides

`spe()`

An effect distribution and average effect, with **bootstrap confidence bounds**.

`ca()`

Tail comparisons, with **bootstrap standard errors and joint p-values**.

SortedEffects 1.7.0. Native SPE inference; audited numerical stabilization for colon CA (details in chapter 05).

Same average, different effect patterns

Constructed example: three equally common baseline profiles.

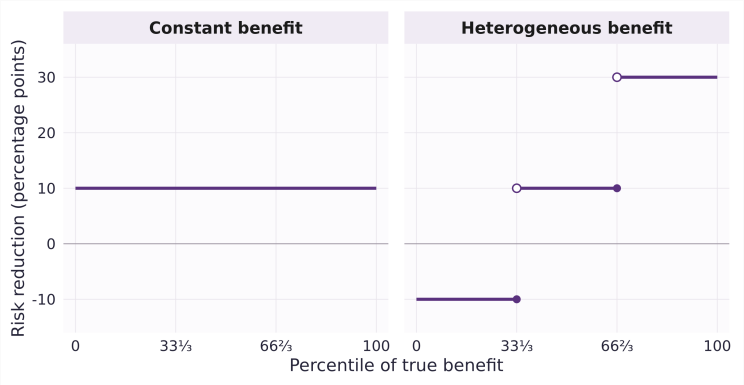
Both populations have **40% control risk**, **30% marginal treated risk** and **10 pp average benefit**.

True benefit (pp)	Profile A	Profile B	Profile C
Constant benefit	+10	+10	+10
Heterogeneous benefit	-10	+10	+30

Benefit = control risk − treated risk.

Constructed conditional risks, not sampled trial outcomes. Control risk is 40% in each profile; treated risks are 30/30/30% or 50/30/10%.

Sorting reveals the difference



- The mean is **10 pp** in both populations.
- Each step spans **one third** of the heterogeneous population.
- The lower third has **negative benefit**, despite a positive average.

In the trial, the effect distribution and its uncertainty must be estimated.

Exact population quantiles of constructed conditional-average effects; no fitted model or sampling uncertainty.

The trial

02

Does chemotherapy reduce mortality by the same amount for every patient profile?

A public colon cancer trial

- Patients with resected **stage III colon cancer**, randomized to three arms.
- We compare **levamisole + 5-FU** with **observation**: 619 patients.
- Outcome: **death within three years**; known for 618 patients.
- Primary analysis: **593 patients** with complete baseline variables.

`survival::colon`, death records; levamisole-only arm excluded. Trial reports (Moertel et al. 1990, 1995).

Start with the average

Observation

34.7%
109 deaths / 314 patients

Levamisole + 5-FU

25.7%
78 deaths / 304 patients

9.1 percentage points lower three-year mortality.

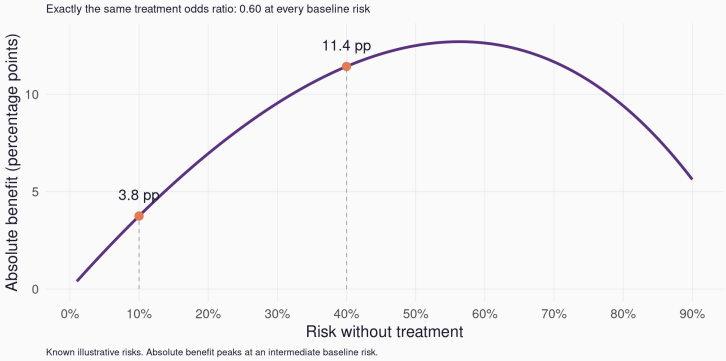
Crude risks: 618 known outcomes. Standardized analyses: 593 complete cases.

Define benefit before sorting

$$\Delta(x) = \Pr(Y = 1 \mid A = 0, X = x) - \Pr(Y = 1 \mid A = 1, X = x).$$

- $Y = 1$: death by three years; $A = 0$: observation; $A = 1$: chemotherapy.
- X: age, sex and tumour characteristics at baseline.
- **Positive:** lower mortality under treatment. **Negative:** higher mortality.
- One conditional **average** contrast per profile, not an individual's causal effect.

Effect scale matters



An algebraic illustration, not an estimate from the colon trial.

- Treatment odds ratio: **0.60** throughout.
- Control risks of **10% and 40%** give reductions of **3.8 and 11.4 pp**.
- Absolute benefit can vary without a treatment interaction on log odds.

Estimated effects

03

Fit the model, predict both conditions, then sort the risk differences.

Specify where effects may differ

```
1  fm <- outcome ~ treatment + splines::ns(age, df = 3) + female +
2    age_z:female + log_nodes + poor_diff + extensive +
3    obstruct + adhere + surg_late +
4    I((treatment == "chemotherapy") * age_z) +
5    I((treatment == "chemotherapy") * female) +
6    I((treatment == "chemotherapy") * age_z * female) +
7    I((treatment == "chemotherapy") * log_nodes) +
8    I((treatment == "chemotherapy") * poor_diff) +
9    I((treatment == "chemotherapy") * extensive)
10
11 fit <- glm(fm, data = dat, family = binomial())
```

1–3: common model · **4–9:** six treatment interactions · **11:** fit the model.

Comparison model: omit lines 4–9. `age_z = (age - 60) / 10`; `log_nodes = log1p(nodes)`.

Predict both treatment conditions

```r
control <- treated <- dat
control$treatment <- "observation"
treated$treatment <- "chemotherapy"

benefit <- predict(fit, control, type = "response") -
  predict(fit, treated, type = "response")
```

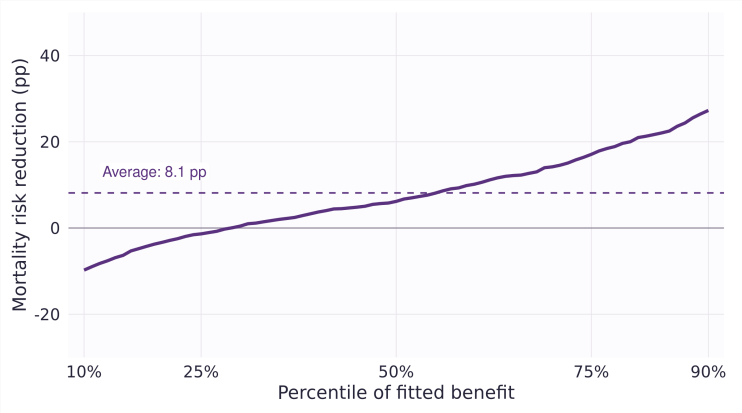
- Keep every patient's baseline characteristics fixed.
- Change only treatment assignment.
- Subtract predicted **risks**, not log odds.

What is being sorted?

$$\mathsf{SPE}(u) = Q_{\Delta(X)}(u), \qquad \mathsf{APE} = E[\Delta(X)].$$

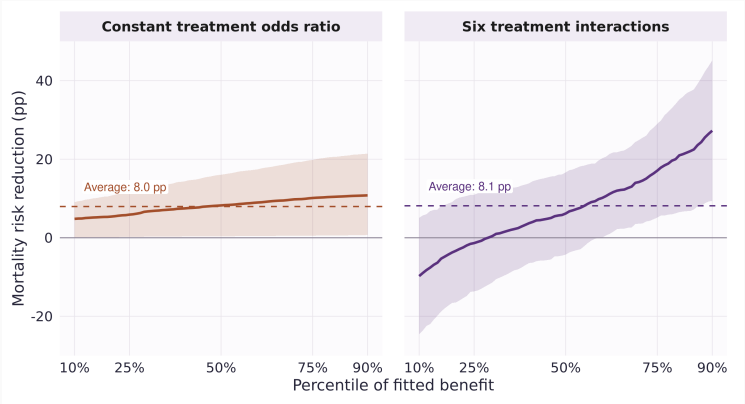
- The distribution of X defines the population summarized: here, 593 observed profiles.
- Classification describes covariates in selected tails of $\Delta(X)$.
- A quantile of effects differs from a difference of outcome quantiles.
- Randomization supports identification; sorting does not replace it.

The sorted-effects curve



- Mean benefit: **8.1 pp**.
- 10th percentile: **−9.8 pp**.
- 90th percentile: **27.3 pp**.
- Each point is a **percentile of fitted benefits**.

Similar averages, different distributions



- Mean benefit: **8.0 vs 8.1 pp**.
- Interactions widen the fitted distribution.
- Six-term interaction test: **p = .014**.
- More spread alone does not validate a ranking.

Native 95% uniform bands over the 10th–90th percentiles, calibrated separately by model.

Classification

04

Which patient profiles sit in the two tails?

Compare the tails with ca()

```r
ca_result <- SortedEffects::ca(
  fm = fm, data = dat,
  method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  u = .10,
  interest = "moment", cl = "diff",
  t = colon_characteristics()$variable,
  b = 499, alpha = .05,
  bc = FALSE, seed = 420
)
```

Model and contrast
Same mortality model; benefit = observation − chemotherapy.

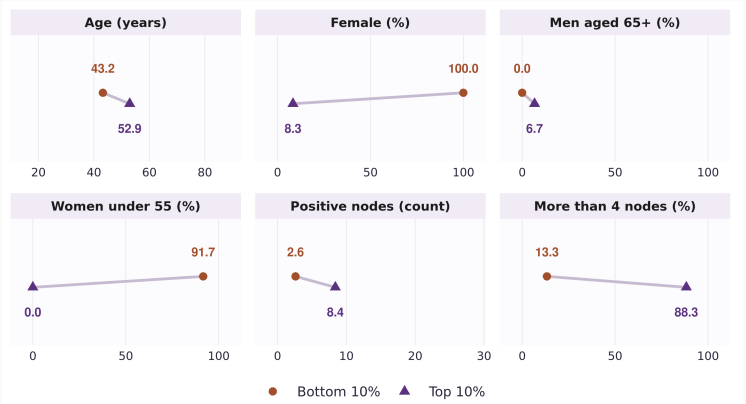
Choose the tails
Bottom and top 10%: 60 patients each here.

Compare characteristics
High − low means, across all twelve traits.

Bootstrap inference
499 draws; SEs and joint p-values.

Colon CA inference uses the audited stabilization of failed native bootstrap fits (chapter 05).

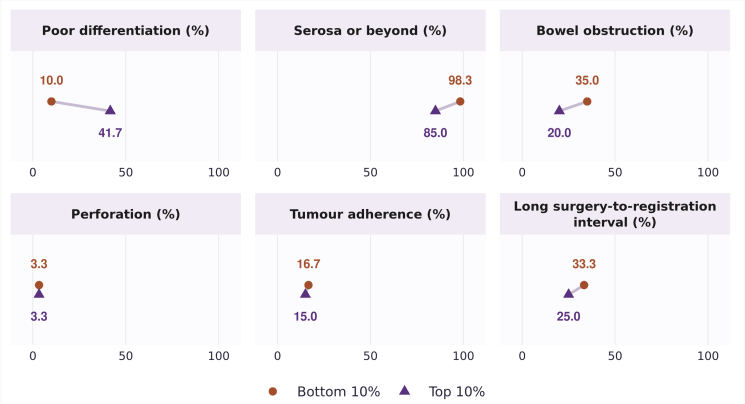
Age, sex and positive nodes



- Low tail: **all women**, mean age **43**.
- High tail: **92% men**, mean age **53**.
- More than four nodes: **13% vs 88%**.
- Men aged 65+: **7% of the high tail**.

Descriptive means; 60 patients per tail. Connecting segments are not confidence intervals.

Other tumour and clinical characteristics



"Adherence" denotes tumour adherence to nearby organs.

- Poor differentiation: **10% vs 42%**.
- Bowel obstruction: **35% vs 20%**.
- Perforation: **3% in both tails**.
- Inference for these differences follows.

Tail composition is not a subgroup treatment effect

- Tail membership depends jointly on **age, sex and tumour characteristics**.
- Younger women dominate the fitted bottom tenth.
- Their representation does not estimate the treatment effect in all women.

Assessing harm requires a treatment contrast and its uncertainty for a defined population.

Inference and validation

05

How much of the pattern could be sampling variation or model choice?

Estimate the curve and its uncertainty

```r
spe_result <- SortedEffects::spe(
  fm = fm, data = dat,
  method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  us = seq(.10, .90, .01),
  b = 499,
  alpha = .05, bc = FALSE,
  seed = 420
)
```

Fit the outcome model
Our chosen formula and a logistic link.
Set the contrast
This ordering gives observation − chemotherapy.
Choose the percentiles
10th–90th, in one-percentile steps.
Quantify uncertainty
499 draws; 95% bounds; no bias correction.

Uniform coverage concerns the chosen percentile grid and requires valid effect estimation and regularity conditions.

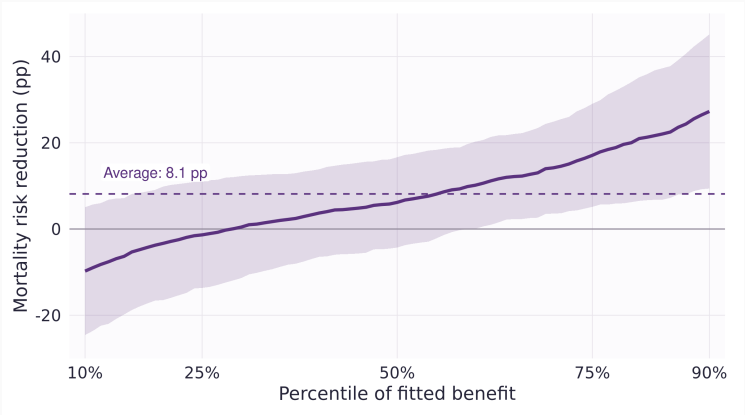
How is the uniform band built?

1. Refit in each of **499** bootstrap draws and re-sort the fitted benefits at 81 percentiles, 10th–90th.
2. Scale each percentile by its bootstrap interquartile range ÷ 1.349.
3. Critical value: 95th percentile of the **largest** scaled deviation across all 81 percentiles.
4. Band: estimate ± critical value × scale; endpoints rearranged to stay monotone.

Model	Uniform critical value	Pointwise
Six treatment interactions	2.79	1.96
Constant treatment odds ratio	2.17	1.96

Native SortedEffects 1.7.0 IQR/max bootstrap, independently reconstructed; critical values recovered from the saved bands and scales. Rearrangement: Chernozhukov, Fernández-Val, and Galichon (2009).

Uncertainty in the lower tail



- 10th percentile: **−9.8 pp**.
- Band at that point: **−24.6 to +5.1 pp**.
- The sign is uncertain at the lower end.
- Mean benefit: **8.1 pp**, CI **1.4 to 14.9**.

Native 95% uniform SPE band; separate 95% interval for the average benefit.

Are the tail characteristics different?

Characteristic	Low tail	High tail	Joint p
Female	100%	8%	.042
Women under 55	92%	0%	.040
More than four nodes	13%	88%	.132
Poor differentiation	10%	42%	.948

- Joint comparisons account for **all twelve specified characteristics**.
- The two small p-values concern closely related traits.

Audited stabilized CA, 499 draws. Full twelve-trait table in backup.

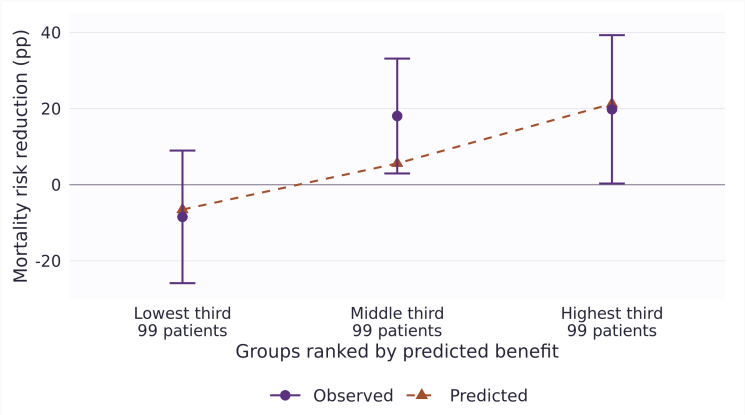
What went wrong in native CA?

- CA rescales bootstrap fitting weights to total **20,000**.
- **141/499 fits did not converge**; 137 draws had undefined tail moments.
- The routine still returned finite-looking inference, which we do not use.
- Stable starting coefficients recovered **499 converged, finite draws**, with none discarded.

Check every bootstrap fit: finite-looking output is not necessarily valid.

Reconstruction preserves resamples, likelihood, threshold definitions and moment formulas.

Does the ranking work in held-out patients?



- Fit on **296**; freeze predictions for **297**.
- One split; three groups of **99**.
- Ranking slope **1.10** (0 = no signal, 1 = calibrated); CI **0.24–1.97**, adjusted **p = .025**.
- Low-third benefit: **−8.5 pp**, CI **−25.9 to +9.0**.

Pointwise, unadjusted 95% intervals in the plot. Slope p-value: Holm adjustment over the two planned rankings.

Robustness and scope

- **Missing-data sensitivity:** a simple fill-and-indicator analysis of all 618 known outcomes changes interaction p from **.014 to .054**.
- **One held-out split:** useful, but not external validation.
- **Small tails and flexible modeling:** 60 patients per tail; 19 parameters for 179 deaths.
- **Exploratory search:** our third trial pilot; no adjustment across datasets.

Exploratory evidence for a ranking; further validation is needed before treatment decisions.

Discussion

06

What would make a sorted-effects analysis useful in our own work?

What did sorting add?

- Two models gave almost the **same average benefit**, but different effect distributions.
- Classification made the tails concrete: age, sex and tumour features together.
- Uncertainty and held-out results changed how strongly we could describe the pattern.

Which result would you want to see before taking this pattern seriously?

Where do sorted effects fit?

Approach	Question	Watch for
Subgroup forest plot	Does the effect differ by one trait?	One trait at a time; many tests
Risk modeling	Does absolute benefit change with baseline risk?	Misses modifiers unrelated to risk
Effect modeling	Which traits modify the effect?	Overfitting; needs validation
Sorted effects + classification	How are fitted effects distributed, and who is in the tails?	Only as good as the effect estimates

Here, the constant-odds-ratio model is risk-based; the six interactions are effect modeling.

Risk and effect modeling as defined in the PATH statement (Kent et al. 2020).

Questions for discussion

- **Spread or noise?** The interaction model gives 29% of patients a negative fitted benefit, with 19 parameters for 179 deaths. Shrink before sorting?
- **Prespecification:** Where would sorted effects belong in a trial's analysis plan: secondary or exploratory?
- **Robustness:** Keeping all 618 patients moves the interaction p-value from .014 to .054. Which analysis should lead?
- **Scale:** Absolute benefit varies even with a constant odds ratio. On which scale should heterogeneity be claimed?

Next time: causal forests

Could the data learn the age–sex–tumour pattern?

- **Our regression:** we choose which interactions to include.
- **Causal forests:** many trees learn nonlinearities and interactions.
- **SPE + classification:** could summarize either set of effect estimates and describe the tails.

Causal in either case: randomization—or no unmeasured confounding given baseline variables—plus overlap and well-defined treatment.

Flexibility changes estimation; causal identification comes from design and assumptions. Generalized random forests (Athey, Tibshirani, and Wager 2019).

How a forest builds a comparison

For a patient profile, which treated and control patients are comparable?

1 · Grow a tree

Split baseline profiles to find groups with different treatment–control contrasts.

2 · Grow many trees

Different samples and candidate splits give different groups of similar patients.

3 · Combine the comparisons

Patients repeatedly grouped with the target profile get more weight in its treatment comparison.

Many trees reduce dependence on any one noisy split.

“Honest” fitting uses separate observations to choose splits and estimate effects. Randomized-trial intuition; no forest was fitted here. (Athey, Tibshirani, and Wager 2019)

Reading and reproducible materials

- **Send ahead:** the R tutorial, introduction and applied example (Chen et al. 2020).
- **Optional theory:** the original method paper (Chernozhukov, Fernández-Val, and Luo 2018).
- **Analysis report:** full classification table, validation, sensitivities and data provenance.
- **Code and slides:** github.com/valentinklotzbuecher/jcod.

Additional material

+

Full tables, causal assumptions and resampling details.

Backup · When is the heterogeneity causal?

- **Identification, both:** randomization or no unmeasured confounding; overlap; consistency/no interference; appropriate handling of selection and missing data.
- **Regression → SPE/CA:** reliable conditional-risk estimates; suitable convergence and regularity of the effect distribution.
- **Forests:** enough local data, regularity, suitable nuisance estimates and honest fitting.
- **Inference:** pointwise forest intervals do not automatically cover sorted effects or selected tails.

Both concern conditional average effects. Fitted spread alone does not establish true heterogeneity. (Chernozhukov, Fernández-Val, and Luo 2018; Athey, Tibshirani, and Wager 2019)

Backup · Classification: demographics and nodes

Characteristic	Low	High	Difference	SE	Joint p
Age (years)	43.2	52.9	+9.7	7.1	0.872
Female (%)	100.0	8.3	-91.7	21.4	0.042
Men aged 65+ (%)	0.0	6.7	+6.7	14.2	1.000
Women under 55 (%)	91.7	0.0	-91.7	21.1	0.040
Positive nodes	2.6	8.4	+5.8	2.2	0.297
More than 4 nodes (%)	13.3	88.3	+75.0	22.1	0.132

Difference = high − low. Percentages differ in percentage points.

60 patients per tail. SEs and joint p-values: audited stabilized CA, 499 draws. Adjustment covers all twelve traits, not just these six.

Backup · Classification: clinical characteristics

Characteristic	Low	High	Difference	SE	Joint p
Poor differentiation (%)	10.0	41.7	+31.7	27.7	0.948
Serosa or beyond (%)	98.3	85.0	-13.3	17.4	0.992
Bowel obstruction (%)	35.0	20.0	-15.0	9.6	0.762
Perforation (%)	3.3	3.3	+0.0	3.7	1.000
Tumour adherence (%)	16.7	15.0	-1.7	10.5	1.000
Long surgery interval (%)	33.3	25.0	-8.3	10.8	0.990

Difference = high − low, in percentage points.

“Long surgery interval”: surgery-to-registration category. SEs and joint p-values: audited stabilized CA, adjusted across all twelve traits.

Backup · Calculate tail means

```r
cutoff <- Hmisc::wtd.quantile(
  benefit, rep(1, nrow(dat)), c(.10, .90)
)
tails <- list(low = benefit <= cutoff[1],
              high = benefit >= cutoff[2])
traits <- colon_characteristics()$variable
lapply(tails, function(i)
  colMeans(dat[i, traits]))
```

- Select tails by fitted benefit, then summarize baseline characteristics.
- Both tails can contain patients from either randomized arm.

Descriptive calculation only. `R/colon-walkthrough.R` verifies these means against the saved classification table.

Backup · What is resampled?

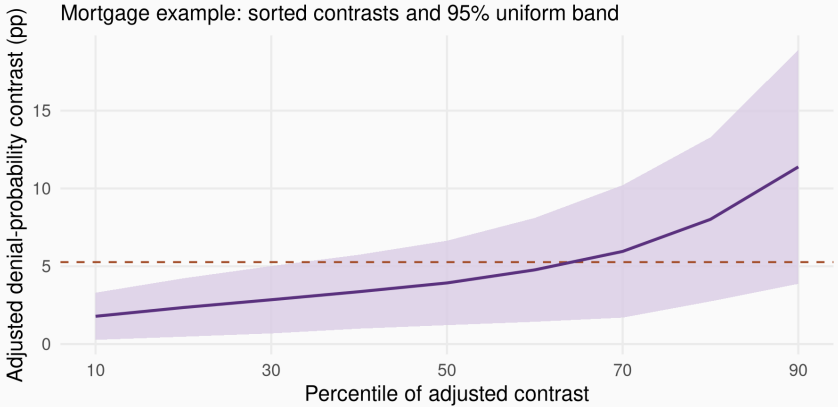
- `spe()` **full sample:** refits with bootstrap weights; effect quantiles retain original profile weights.
- `ca()` **moments:** refits and updates thresholds and characteristic means with resampled weights.
- **Held-out validation:** fits on training data; evaluates the frozen ranking using test outcomes.

Backup · Cohort and sensitivity checks

Analysis	Patients	Average benefit	Interaction p
Complete baseline and endpoint	593	8.1 pp	.014
Missing baseline values with indicators	618	8.4 pp	.054
Early-censored outcome assigned 0	594	8.0 pp	.017
Early-censored outcome assigned 1	594	8.4 pp	.010

- The latter two retain complete baseline variables.
- Eight records disagree on the supplied vs count-derived four-node indicator; the report includes a coding check.

Backup · Original published example



Tutorial benchmark: adjusted observational disparity (Munnell et al. 1996; Chen et al. 2020).

References I

Athey, Susan, Julie Tibshirani, and Stefan Wager, "Generalized random forests," *The Annals of Statistics*, 47 (2019), 1148–1178.

Chen, Shuowen, Victor Chernozhukov, Iván Fernández-Val, and Ye Luo, "SortedEffects: Sorted causal effects in R," *The R Journal*, 12 (2020), 131–146.

Chernozhukov, Victor, Iván Fernández-Val, and Alfred Galichon, "Improving point and interval estimators of monotone functions by rearrangement," *Biometrika*, 96 (2009), 559–575.

Chernozhukov, Victor, Iván Fernández-Val, and Ye Luo, "The sorted effects method: Discovering heterogeneous effects beyond their averages," *Econometrica*, 86 (2018), 1911–1938.

Kent, David M., Jessica K. Paulus, David van Klaveren, and others, "The predictive approaches to treatment effect heterogeneity (PATH) statement," *Annals of Internal Medicine*, 172 (2020), 35–45.

Moertel, Charles G., Thomas R. Fleming, John S. Macdonald, and others, "Levamisole and fluorouracil for adjuvant therapy of resected colon carcinoma," *New England Journal of Medicine*, 322 (1990), 352–358.

——, "Fluorouracil plus levamisole as effective adjuvant therapy after resection of stage III colon carcinoma: A final report," *Annals of Internal Medicine*, 122 (1995), 321–326.

References II

Munnell, Alicia H., Geoffrey M. B. Tootell, Lynn E. Browne, and James McEneaney, “Mortgage lending in Boston: Interpreting HMDA data,” *The American Economic Review*, 86 (1996), 25–53.

van Klaveren, David, Ewout W. Steyerberg, Patrick W. Serruys, and David M. Kent, “The proposed ‘concordance-statistic for benefit’ provided a useful metric when modeling heterogeneous treatment effects,” *Journal of Clinical Epidemiology*, 94 (2018), 59–68.